BIO 304 . HUMAN ANATOMY & PHYSIOLOGY . WEEK 2 . MODULE 4 OF 16

Module 4: Integumentary System

Topics: Skin Structure and Layers, Skin Functions and Accessory Structures

TOPIC 1 OF 2 . WEEK 2 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

Skin Structure and Layers

Epidermis, dermis, hypodermis: who lives where and what they do.

The Science

Skin is your largest organ by surface area. Two layers strictly (epidermis, dermis) plus a hypodermis underneath that is technically not skin but is functionally part of the system. Each layer has a job.

Epidermis: stratified squamous keratinized, avascular, several sub-layers (basale, spinosum, granulosum, lucidum in thick skin only, corneum). Keratinocytes are born at the basale, mature upward, die, and form the cornified layer that protects against everything from microbes to water loss. Melanocytes in the basale produce melanin and transfer it to keratinocytes for UV protection. Langerhans cells (immune surveillance) and Merkel cells (touch) live here too.

Dermis: vascular, where the action is. Papillary layer has capillary loops and fine sensory endings; reticular layer is dense irregular connective tissue, the source of skin's strength. Hair follicles, glands, and nerves are dermal. Hypodermis is adipose and loose connective tissue, providing insulation and energy storage.

Teaching This Topic

Before the video (drawing prompt)

Have students draw a cross-section of skin and try to label five layers/structures before watching.

Common misconceptions to address

- Students think the epidermis is vascular. It is avascular; it lives off diffusion from the dermis.
- Tanning is sometimes seen as healthy. Tanning is the keratinocyte response to UV damage; melanin is protective but the damage has already occurred.
- The hypodermis is often called part of the skin. Technically it is not, but functionally it is inseparable.

Order of operations (what to teach first)

Three layers overview, then epidermis with sub-layers, then dermis with vascular and sensory content, then hypodermis. Build from outside in.

Self-test prompts

- List the layers of the epidermis from deep to superficial in thick skin.
- Why is the epidermis avascular?
- Where in the skin would a needle for an intradermal injection actually deliver the medication?

Why This Matters to Your Patients

Burn depth: first-degree burns are epidermal (sunburn-like, painful, no scar). Second-degree are partial-thickness (blister, painful, may scar). Third-degree are full-thickness (extending into or through the dermis; the tissue from which new skin would regrow is gone, requiring grafting). Pressure ulcers, IV infiltrations, melanoma screening, transdermal drug delivery, all of these depend on knowing exactly where in the skin you are. A nurse who can name skin layers reads burns, wounds, and skin lesions with confidence.

TOPIC 2 OF 2 . WEEK 2 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

Skin Functions and Accessory Structures

Protection, thermoregulation, sensation, vitamin D; plus hair, glands, and nails.

The Science

Skin functions are not a list to memorize, they are a checklist for clinical assessment. Protection (barrier, immunity), thermoregulation (sweat and vasomotor), sensation (touch, pressure, temperature, pain), vitamin D synthesis (UVB on 7-dehydrocholesterol), and minor excretion (small amounts of nitrogen waste in sweat). Lose any of these and the patient suffers.

Accessory structures: hair (insulation, sensation, identification), nails (protection of digit tips, fine manipulation), eccrine sweat glands (thermoregulation), apocrine sweat glands (axilla and groin, activated at puberty, produce odor when bacteria feed on the secretion), sebaceous glands (oily sebum, waterproofs, slightly antimicrobial). Each accessory structure is a specialization of skin tissue.

Thermoregulation deserves a beat: vasodilation brings warm blood to the skin surface for heat dissipation (radiation, conduction), sweating cools by evaporation (water's high heat of vaporization, see Chemistry of Life), vasoconstriction conserves core heat. The countercurrent arrangement between cutaneous arteries and veins is an additional layer of thermal control.

Teaching This Topic

Before the video (drawing prompt)

Ask students to list every function of skin they can think of in 60 seconds, then compare to your list. Most will get fewer than half.

Common misconceptions to address

- Skin is seen as cosmetic. It is the body's largest immune and thermoregulatory organ.
- Sweat is thought to cool by 'feeling cool.' It cools by evaporating; if it just drips off without evaporating, no cooling occurs.
- Vitamin D from skin is often overlooked. Sun exposure is a major route; deficiency is common in dark-skinned people at high latitudes.

Order of operations (what to teach first)

Functions first (with thermoregulation expanded), then accessory structures and how each contributes.

Self-test prompts

- Name five functions of the integumentary system.
- Explain how sweating actually cools the body.
- Why are dark-skinned people at higher risk of vitamin D deficiency in northern climates?

Why This Matters to Your Patients

Burn patients lose fluid through damaged skin at staggering rates and die of dehydration, hypothermia, and infection long before the burn itself would kill them. Diabetics with poor circulation develop pressure ulcers because skin cannot defend or repair itself without good blood flow. Patients with autonomic dysfunction lose thermoregulation; spinal cord injury patients above T6 cannot vasoconstrict below the level of injury and are at risk for hypothermia in mild cold. Vitamin D status affects bone health, immunity, and probably mood; supplementation matters. Skin assessment is the most accessible window into systemic disease the nurse has.